Air Quality Classification

Using Environmental and Pollution Indicators

USING LOGISTIC REGRESSION

Team no 49

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View project on GitHub

PROBLEM AND DATA

WHAT AND WHY

- Air pollution affects human health and the environment, making air quality monitoring essential.
- Environmental and pollution indicators help identify and analyse different levels of air pollution.
- Logistic Regression helps classify air quality into Good, Moderate, Poor and Hazardous categories.

DATASET

- Kaggle: Air Quality and Pollution Assessment
- Total rows: 5000
- Total columns: 10

COLUMNS

- Input columns:

Temperature, Humidity, PM2.5, PM10, NO2, SO2, CO, Population_Density Proximity_to_Industrial_Areas

- Output column: Air Quality

APPROACH

DATA

- Load CSV, clean data and remove duplicates
- 9 input features: Temperature, SO2, Humidity,PM2.5, NO2,PM10 CO,Population_Density Proximity_to_Industrial_Areas
- Target: Air quality

MODEL

- 80/20 Train-Test Split with data processing
- Logistic Regression and Random Forest (Scikit-learn)

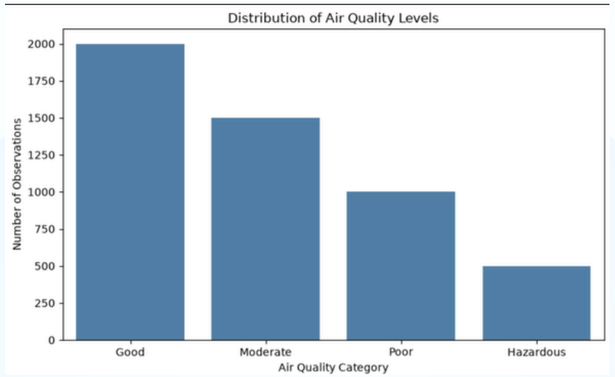
OUTPUT

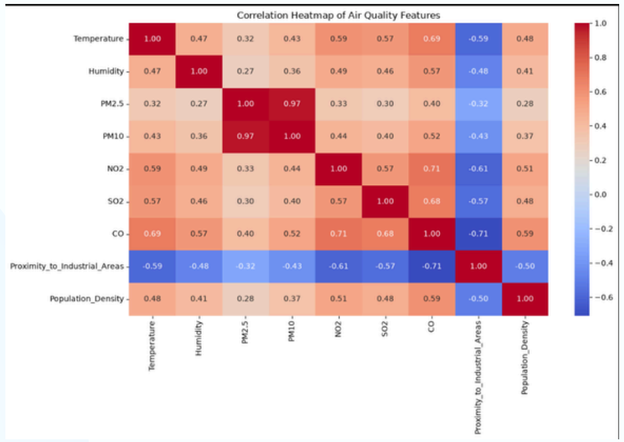
- Predicted Air Quality: Good, Moderate, Poor, Hazardous
- Accuracy, Confusion Matrix, Classification Report and Feature Importance

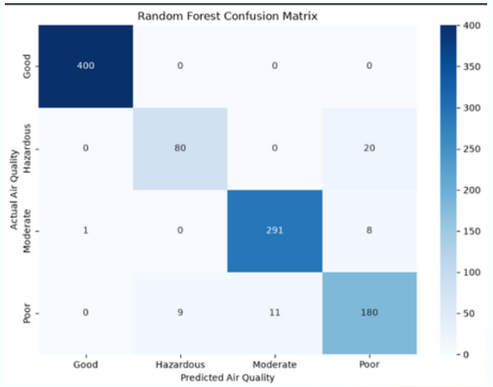
LIBRARIES USED

pandas numpy matplotlib seaborn scikit-learn

RESULTS








```
Logistic Regression Accuracy: 92.9 %
Random Forest Accuracy: 95.1 %

MODEL COMPARISON
Logistic Regression: 92.9 %
Random Forest: 95.1 %

Best Model: Random Forest
LOGISTIC REGRESSION REPORT
              precision    recall  f1-score   support

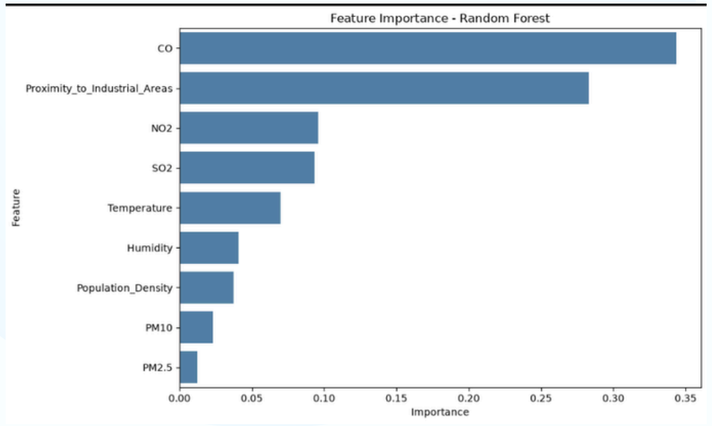
        Good       0.99      1.00      0.99       400
   Hazardous       0.90      0.77      0.83       100
    Moderate       0.93      0.94      0.94       300
        Poor       0.82      0.85      0.84       200

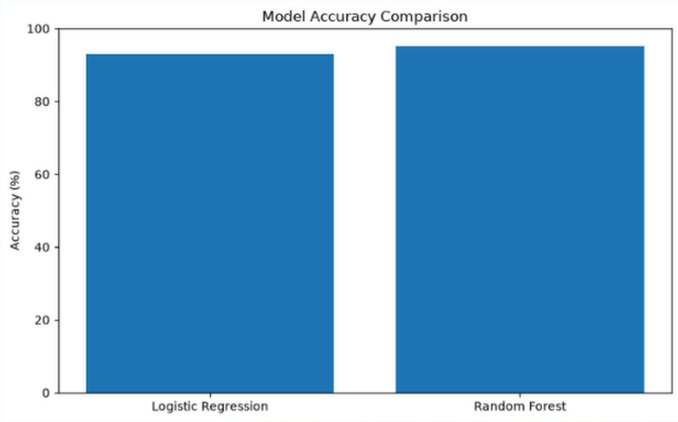
    accuracy                           0.93      1000
   macro avg       0.91      0.89      0.90      1000
weighted avg       0.93      0.93      0.93      1000


RANDOM FOREST REPORT
              precision    recall  f1-score   support

        Good       1.00      1.00      1.00       400
   Hazardous       0.90      0.80      0.85       100
    Moderate       0.96      0.97      0.97       300
        Poor       0.87      0.90      0.88       200

    accuracy                           0.95      1000
   macro avg       0.93      0.92      0.92      1000
weighted avg       0.95      0.95      0.95      1000
```





Random Forest Accuracy: 95.1%

Logistic Regression Accuracy: 92.9%

Input: Environmental and pollution indicators

Predicted Air Quality: GOOD

Random Forest achieved better classification accuracy than Logistic Regression, making it the best-performing model for this dataset.

LEARNINGS

WHAT WORKED

- Both logistic regression and Random Forest successfully classify air quality levels
- Feature importance helps identify which pollution factors influence air quality the most

WHAT DIDN'T WORK

- Logistic Regression may miss complex, non-linear relationships between pollution factors
- Missing or non-numeric values require extra data cleaning and preprocessing

AI-use disclosure: AI tool (ChatGPT) was used for code assistance

Future Scope: Develop a real-time air quality prediction system using live sensor data to improve pollution monitoring and public awareness

ANKITA	DIVIJA	SHIVANSH	PRANJAL
Documentation Lead: README and report	PPT Lead: Made presentation	Video Lead: Demo video	Execution Lead: Ran the code and results